Generalized Order Statistics

Let F be a continuous cumulative distribution function with corresponding quantile function F^{-1}, and let $(\gamma_i)_{i\geq 1}$ be a sequence of positive parameters. Let $U_1, U_2, \ldots$ be independent random variables, each uniformly distributed on $(0,1)$.

For each $r \geq 1$, define

$$U_{r,\tilde{\gamma}_r} := 1 - \prod_{i=1}^{r} U_i^{1/\gamma_i}, \qquad \tilde{\gamma}_r := (\gamma_1, \ldots, \gamma_r).$$

The sequence $\left(U_{r,\tilde{\gamma}_r}\right)_{r\geq 1}$ is referred to as a sequence of *uniform generalized order statistics* (U-GOS) associated with the parameter sequence $(\gamma_i)_{i\geq 1}$.

The corresponding sequence of *generalized order statistics* associated with F is defined by applying the quantile function F^{-1} to the uniform GOS. Specifically, for $r \geq 1$, let

$$X_{r,\tilde{\gamma}_r} := F^{-1}(U_{r,\tilde{\gamma}_r}) = F^{-1}\left(1 - \prod_{i=1}^{r} U_i^{1/\gamma_i}\right). \tag{1}$$

Thus, $X_{r,\tilde{\gamma}_r}$ has the distribution induced by F^{-1} acting on the corresponding uniform GOS.

Basic Properties

- The sequence $(X_{r,\tilde{\gamma}_r})_{r\geq 1}$ is almost surely increasing:

  $$X_{1,\tilde{\gamma}_1} \leq X_{2,\tilde{\gamma}_2} \leq \cdots$$

  since the product $\prod_{i=1}^{r} U_i^{1/\gamma_i}$ is decreasing in r.

- The distribution of $X_{r,\tilde{\gamma}_r}$ depends only on the first r parameters $(\gamma_1, \ldots, \gamma_r)$.

- The construction is invariant under permutations of $(\gamma_1, \ldots, \gamma_r)$ at the marginal level.

Special Cases

- **Ordinary order statistics:** If $\gamma_r = n + 1 - r$, $r = 1, \ldots, n$, then

  $$X_{r,\tilde{\gamma}_r} \equiv X_{r:n},$$

  the r-th order statistic of an i.i.d. sample of size n.

- **k-th upper record values:** If $\gamma_i = k$ for all $i \geq 1$, then $(X_{r,\tilde{\gamma}_r})$ corresponds to the sequence of k-th upper record values. In particular, $k = 1$ yields classical record values.

- **Progressively censored order statistics:** For a censoring scheme $R = (R_1, \ldots, R_n)$, the model corresponds to generalized order statistics with parameters

  $$\gamma_r = \sum_{i=r}^{n} (R_i + 1), \quad r = 1, \ldots, n.$$

The classes $\mathcal{F}_G^{cx}$ and $\mathcal{F}_G^{cv}$ can be characterized in terms of the expectation of the generalized order statistic (GOS), $\mathbb{E}[X_{r,\tilde{\gamma}_r}]$. To make this characterization explicit, let

$$U_{r,\tilde{\gamma}_r} = 1 - \prod_{i=1}^{r} U_i^{1/\gamma_i},$$

where $U_1, \ldots, U_r$ are independent random variables uniformly distributed on $(0,1)$, and define

$$\mu_{r,\tilde{\gamma}_r} := \mathbb{E}[X_{r,\tilde{\gamma}_r}], \qquad \pi^G_{r,\tilde{\gamma}_r} := G\big(\mathbb{E}\big[G^{-1}(U_{r,\tilde{\gamma}_r})\big]\big)\,.$$

Assume that the relevant expectations exist and that $G^{-1} \circ F$ is convex. By Jensen's inequality,

$$\begin{aligned}(G^{-1} \circ F)\big(\mu_{r,\tilde{\gamma}_r}\big) &\le \mathbb{E}\big[(G^{-1} \circ F)(X_{r,\tilde{\gamma}_r})\big] \\ &= \mathbb{E}\big[G^{-1}(U_{r,\tilde{\gamma}_r})\big]\,.\end{aligned}$$

Applying G to both sides and using the monotonicity of G, we obtain

$$F\big(\mu_{r,\tilde{\gamma}_r}\big) \le G\big(\mathbb{E}\big[G^{-1}(U_{r,\tilde{\gamma}_r})\big]\big) = \pi^G_{r,\tilde{\gamma}_r}.$$

Consequently, the resulting bound depends only on G, r, and the parameter vector $\tilde{\gamma}_r$, rather than on the particular distribution F. The analogous argument for concave $G^{-1} \circ F$ yields the reverse inequality. These observations lead to the following characterization.

Proposition 1. *Let*

$$\pi^G_{r,\tilde{\gamma}_r} := G\big(\mathbb{E}\big[G^{-1}(U_{r,\tilde{\gamma}_r})\big]\big)\,,$$

and suppose that the expectations involved are well-defined. Then, for every admissible r and parameter vector $\tilde{\gamma}_r$,

1. *If $F \in \mathcal{F}_G^{cx}$, then*
$$\mathbb{P}(X \le \mu_{r,\tilde{\gamma}_r}) \le \pi^G_{r,\tilde{\gamma}_r}.$$
2. *If $F \in \mathcal{F}_G^{cv}$, then*
$$\mathbb{P}(X \le \mu_{r,\tilde{\gamma}_r}) \ge \pi^G_{r,\tilde{\gamma}_r}.$$

1. Estimating the Expected Generalized Order Statistics

Recall from (1) that the generalized order statistic $X_{r,\tilde{\gamma}_r}$ is defined by

$$X_{r,\tilde{\gamma}_r} = F^{-1}(U_{r,\tilde{\gamma}_r})\,,$$

where $U_{r,\tilde{\gamma}_r}$ denotes the corresponding uniform generalized order statistic. Accordingly, its distribution function can be expressed as

$$F_{X_{r,\tilde{\gamma}_r}}(x) = F_{U_{r,\tilde{\gamma}_r}}(F(x))\,, \qquad x \in \mathbb{R}. \tag{2}$$

This representation allows us to derive the distribution of $X_{r,\tilde{\gamma}_r}$ directly from the distribution of its uniform counterpart.

The goal now is to estimate the expected generalized order statistic $\mu_{r,\tilde{\gamma}_r} := \mathbb{E}[X_{r,\tilde{\gamma}_r}]$. To facilitate this estimation, we first obtain explicit expressions for the distribution function in (2). We consider two important cases, according to the structure of the parameter vector $\tilde{\gamma}_r = (\gamma_1, \ldots, \gamma_r)$: **Distinct parameters:** The parameters $\gamma_1, \ldots, \gamma_r$ are pairwise distinct and **Equal parameters:** All parameters coincide, so that $\gamma_1 = \cdots = \gamma_r = k$. The latter case corresponds to upper record values under the associated parameterization. The following two lemmas establish the explicit distribution functions for these cases.

Lemma 2. *suppose that $\gamma_1, \ldots, \gamma_r$ are pairwise distinct positive real numbers. Then:*

- *The distribution function of $U_{r,\tilde{\gamma}_r}$ is given by*

$$F_{U_{r,\tilde{\gamma}_r}}(u) = \begin{cases} 0, & u \le 0, \\ 1 - \sum_{i=1}^{r} \left(\prod_{\substack{j=1 \\ j \neq i}}^{r} \frac{\gamma_j}{\gamma_j - \gamma_i} \right) (1-u)^{\gamma_i}, & 0 < u < 1, \\ 1, & u \ge 1. \end{cases}$$

- *The probability density function of $U_{r,\tilde{\gamma}_r}$ is given by*

$$f_{U_{r,\tilde{\gamma}_r}}(u) = \begin{cases} \sum_{i=1}^{r} \gamma_i \left(\prod_{\substack{j=1 \\ j \neq i}}^{r} \frac{\gamma_j}{\gamma_j - \gamma_i} \right) (1-u)^{\gamma_i - 1}, & 0 < u < 1, \\ 0, & \textit{otherwise.} \end{cases}$$

Lemma 3.

- *The distribution function of the k-th uniform record value $U_{r,\tilde{\gamma}_r} = 1 - \prod_{j=1}^{r} U_j^{1/k}$ is given by: $F_{U_{r,\tilde{\gamma}_r}}(u) = F_{\mathrm{Gamma}(r,1)}(-k \ln(1-u))$.*
- *The density $f_{U_{r,\tilde{\gamma}_r}}$ is bounded if and only if $k \ge 2$ or $(k = 1,\ r = 1)$.*

‘‘‘latex

1.1. A Plug-in Estimator of the Expected Generalized Order Statistic

Let $X_1, \ldots, X_n$ be an i.i.d. sample from a distribution function F, and let F_n denote the corresponding empirical distribution function,

$$F_n(x) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\{X_i \le x\}, \qquad x \in \mathbb{R}.$$

Throughout, we use F_n to denote the empirical distribution function associated with the observed sample, whereas $\mathbb{F}_n$ may be used when emphasizing its random nature.

Recall that the distribution function of the generalized order statistic $X_{r,\tilde{\gamma}_r}$ is given by

$$F_{X_{r,\tilde{\gamma}_r}}(x) = \left(F_{U_{r,\tilde{\gamma}_r}} \circ F\right)(x),$$

where $F_{U_{r,\tilde{\gamma}_r}}$ is the distribution function of the uniform generalized order statistic

$$U_{r,\tilde{\gamma}_r} \stackrel{d}{=} 1 - \prod_{j=1}^{r} U_j^{1/\gamma_j},$$

with $U_1, \ldots, U_r$ independent and uniformly distributed on $(0,1)$, and $\tilde{\gamma}_r = (\gamma_1, \ldots, \gamma_r)$.

Replacing the unknown distribution function F with its empirical counterpart F_n yields the plug-in estimator of the distribution function of $X_{r,\tilde{\gamma}_r}$:

$$\widehat{F}_{X_{r,\tilde{\gamma}_r}}(x) := F_{U_{r,\tilde{\gamma}_r}}(F_n(x))\,.$$

Consequently, provided the expectation exists, the expected generalized order statistic can be written as

$$\mu_{r,\tilde{\gamma}_r}(F) := \mathbb{E}[X_{r,\tilde{\gamma}_r}] = \int_{\mathbb{R}} x\, dF_{X_{r,\tilde{\gamma}_r}}(x).$$

This representation motivates the following plug-in estimator:

$$\begin{aligned}\widehat{\mu}_{r,\tilde{\gamma}_r} &:= \mu_{r,\tilde{\gamma}_r}(F_n) \\ &= \int_{\mathbb{R}} x\, dF_{U_{r,\tilde{\gamma}_r}}(F_n(x))\,.\end{aligned}$$

To obtain an explicit expression, let

$$X_{(1:n)} \le X_{(2:n)} \le \cdots \le X_{(n:n)}$$

denote the sample order statistics. Since the empirical distribution function increases in increments of $1/n$, the composed distribution function $F_{U_{r,\tilde{\gamma}_r}} \circ F_n$ assigns the weight

$$w_{i,n}^{(r,\tilde{\gamma}_r)} := F_{U_{r,\tilde{\gamma}_r}}\left(\frac{i}{n}\right) - F_{U_{r,\tilde{\gamma}_r}}\left(\frac{i-1}{n}\right)$$

to the i-th sample order statistic. It follows that

$$\boxed{\widehat{\mu}_{r,\tilde{\gamma}_r} = \sum_{i=1}^{n} X_{(i:n)} \left[F_{U_{r,\tilde{\gamma}_r}}\left(\frac{i}{n}\right) - F_{U_{r,\tilde{\gamma}_r}}\left(\frac{i-1}{n}\right)\right].} \tag{3}$$

Equivalently, the weights admit the probabilistic representation

$$w_{i,n}^{(r,\tilde{\gamma}_r)} = \mathbb{P}\left(\frac{i-1}{n} < U_{r,\tilde{\gamma}_r} \le \frac{i}{n}\right).$$

Since these weights are nonnegative and sum to one, the estimator $\widehat{\mu}_{r,\tilde{\gamma}_r}$ is a weighted average of the sample order statistics.

In particular, $\widehat{\mu}_{r,\tilde{\gamma}_r}$ is an L-estimator, that is, a linear combination of order statistics whose weights are determined by the distribution of the corresponding uniform generalized order statistic. This representation is especially useful in practice because the weights depend only on r, $\tilde{\gamma}_r$, and the sample size n, and can therefore be computed without knowing the underlying distribution function F. "‘

2. The Tests

Our testing procedure is motivated by the inequalities established in Proposition 1. Given the distribution function G, the order r, and the parameter vector $\tilde{\gamma}_r$, the theoretical bound $\pi^G_{r,\tilde{\gamma}_r}$ can be computed directly. This suggests a natural strategy: estimate the left-hand side of the corresponding inequality from the data and compare it with this theoretical bound.

Let $X_1, \ldots, X_n$ be a random sample from the distribution F. We first estimate the expected generalized order statistic

$$\mu_{r,\tilde{\gamma}_r} = \mathbb{E}[X_{r,\tilde{\gamma}_r}]$$

by an estimator $\widehat{\mu}_{r,\tilde{\gamma}_r}$. Here, the sample size n and the GOS index r are not necessarily equal. We then estimate the probability $F(\mu_{r,\tilde{\gamma}_r})$ using the empirical distribution function evaluated at the estimated threshold:

$$\widehat{F}_n(\widehat{\mu}_{r,\tilde{\gamma}_r}) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{X_i \le \widehat{\mu}_{r,\tilde{\gamma}_r}\}\,.$$

Finally, we compare this empirical estimate with the deterministic $\pi^G_{r,\tilde{\gamma}_r}$. To be more concret, let the following hypotheses

To simplify notation, define the vectors

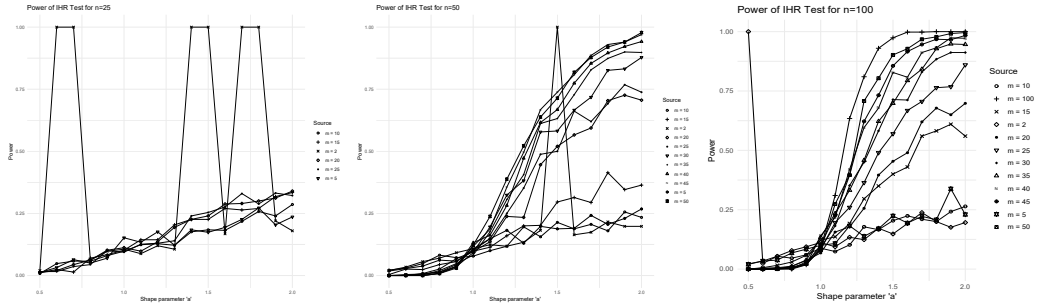
$$\hat{\mathbf{V}}_r = (F_n(\widehat{\mu}_{1,\tilde{\gamma}_1}), \ldots, F_n(\widehat{\mu}_{r,\tilde{\gamma}_r}))\,, \quad \text{and} \quad \mathbf{v}^G_r = \left(\pi^G_{1,\tilde{\gamma}_1}, \ldots, \pi^G_{r,\tilde{\gamma}_r}\right).$$

Thus, the test statistic for $\mathcal{H}^G_{1+}$ is given by

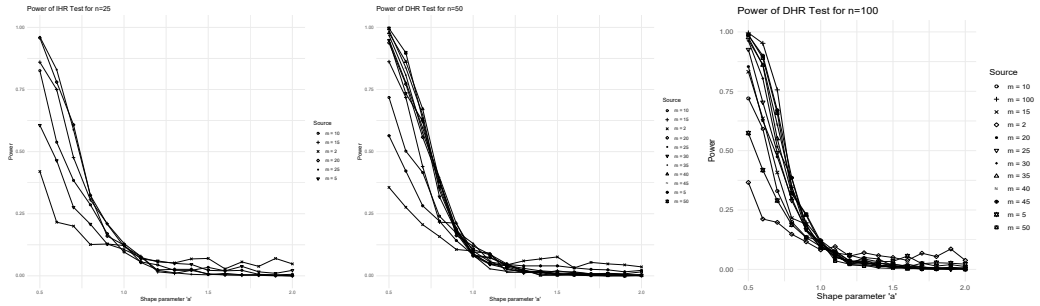
$$T^{G+}_{r,\tilde{\gamma}_r,p}(F_n) = \left\| \left(\mathbf{v}^G_r - \hat{\mathbf{V}}_r\right)_+ \right\|_p = \left(\sum_{k=1}^{r} \left(\pi^G_{k,\tilde{\gamma}_k} - F_n(\hat{\mu}_{k,\tilde{\gamma}_k})\right)_+^p \right)^{1/p}.$$

Similarly, to test $\mathcal{H}^G_{1-}$, we define

$$T^{G-}_{r\tilde{\gamma}_r,p}(F_n) = \left\| \left(\mathbf{v}^G_r - \hat{\mathbf{V}}_r\right)_- \right\|_p.$$



(a) $W(0.75, 1)$.



(b) $B(0.75, 1)$

Figure 1: Simulation results: Rejection rates of the test based on order statistics for the Weibull distribution $W(a, 1)$, with shape parameter $a \in [0.5, 2]$ and sample sizes $n \in \{25, 50, 100\}$.

3. Simulations

4. Proofs

Proof of Lemma 2. Let

$$E_j = -\log U_j, \qquad j = 1, \ldots, r.$$

Since $U_j \sim \text{Uniform}(0, 1)$, the random variables $E_1, \ldots, E_r$ are independent and exponentially distributed with rate 1. Consequently, E_j/γ_j is exponentially distributed with rate γ_j.

Define

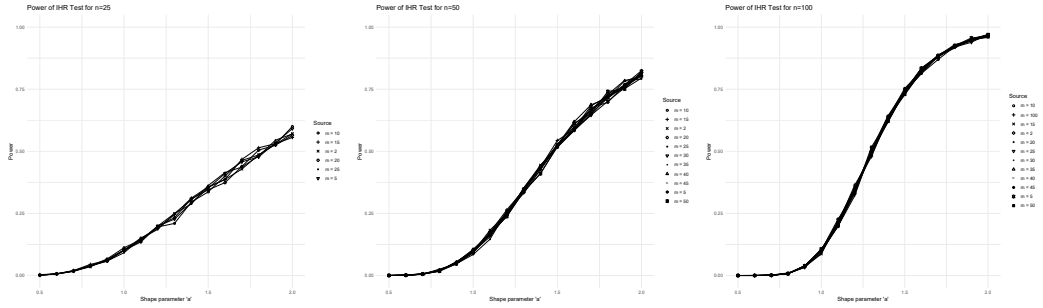
$$S_r = \sum_{j=1}^{r} \frac{E_j}{\gamma_j}.$$

By the definition of $U_{r,\tilde{\gamma}_r}$,

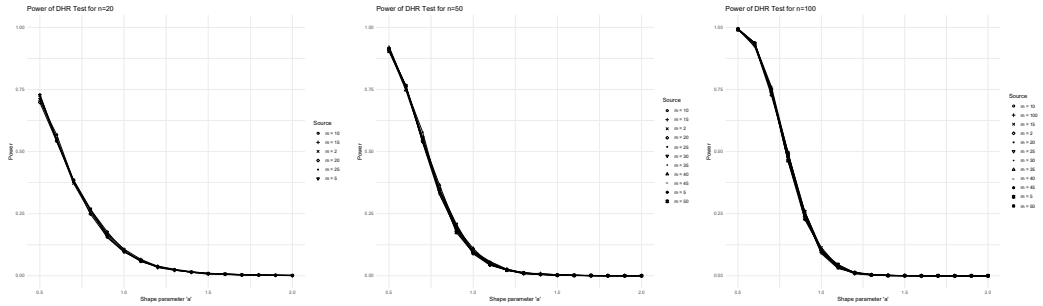
$$U_{r,\tilde{\gamma}_r} = 1 - e^{-S_r}.$$

Therefore, for $0 < u < 1$,

$$\begin{aligned} F_{U_{r,\tilde{\gamma}_r}}(u) &= \mathbb{P}(1 - e^{-S_r} \le u) \\ &= \mathbb{P}\big(S_r \le -\log(1-u)\big) \\ &= 1 - \mathbb{P}\big(S_r > -\log(1-u)\big). \end{aligned}$$



(a) IHR.



(b) DHR.

Figure 2: Simulation results: Rejection rates of the test based on upper record values for the Weibull distribution $W(a, 1)$, with shape parameter $a \in [0.5, 2]$ and sample sizes $n \in \{25, 50, 100\}$.

The Laplace transform of S_r is

$$\mathbb{E}[e^{-sS_r}] = \prod_{j=1}^{r} \frac{\gamma_j}{\gamma_j + s}, \qquad s \geq 0.$$

Since the rates $\gamma_1, \ldots, \gamma_r$ are pairwise distinct, partial-fraction decomposition gives

$$\prod_{j=1}^{r} \frac{\gamma_j}{\gamma_j + s} = \sum_{i=1}^{r} \frac{\gamma_i}{\gamma_i + s} \prod_{\substack{j=1 \\ j \neq i}}^{r} \frac{\gamma_j}{\gamma_j - \gamma_i}.$$

It follows that the survival function of S_r is

$$\mathbb{P}(S_r > t) = \sum_{i=1}^{r} \left(\prod_{\substack{j=1 \\ j \neq i}}^{r} \frac{\gamma_j}{\gamma_j - \gamma_i} \right) e^{-\gamma_i t}, \qquad t \geq 0.$$

Substituting $t = -\log(1-u)$, we obtain

$$\begin{aligned} F_{U_{r,\tilde{\gamma}_r}}(u) &= 1 - \sum_{i=1}^{r} \left(\prod_{\substack{j=1 \\ j \neq i}}^{r} \frac{\gamma_j}{\gamma_j - \gamma_i} \right) e^{\gamma_i \log(1-u)} \\ &= 1 - \sum_{i=1}^{r} \left(\prod_{\substack{j=1 \\ j \neq i}}^{r} \frac{\gamma_j}{\gamma_j - \gamma_i} \right) (1-u)^{\gamma_i}. \end{aligned}$$

Finally, $U_{r,\tilde{\gamma}_r} \in (0,1)$ almost surely, so its CDF equals 0 for $u \leq 0$ and 1 for $u \geq 1$. This establishes the result. □

Proof of Lemma 3.

- The CDF can be expressed as:

$$\begin{aligned} F_{U_{r,\tilde{\gamma}_r}}(u) &= P\left(1 - \prod_{j=1}^{r} U_j^{1/k} \leq u\right) \\ &= P\left(\prod_{j=1}^{r} U_j \geq (1-u)^k\right) \\ &= P\left(-\sum_{j=1}^{r} \ln U_j \leq -k\ln(1-u)\right). \end{aligned}$$

  Since $U_1, \ldots, U_r$ are i.i.d. Uniform$(0,1)$ random variables, the transformed random variables $-\ln U_1, \ldots, -\ln U_r$ are independent and each follows an exponential distribution with rate 1. Consequently, the sum is of Gamma$(r,1)$ distribution: $-\sum_{j=1}^{r} \ln U_j \sim \text{Gamma}(r,1)$.

- Differentiating the CDF with respect to u gives the PDF.

$$f_{U_{r,\tilde{\gamma}_r}}(u) = \begin{cases} \dfrac{k^r}{(r-1)!}(1-u)^{k-1}\big[-\ln(1-u)\big]^{r-1}, & 0 < u < 1, \\ 0, & \text{otherwise.} \end{cases}$$

  Since $f_{U_{r,\tilde{\gamma}_r}}$ is continuous on $(0,1)$, its boundedness can be determined by examining its behavior at the endpoints. As $r \geq 1$, the density satisfies

$$\lim_{u\to 0^+} f_{U_{r,\tilde{\gamma}_r}}(u) = \begin{cases} k, & r = 1, \\ 0, & r > 1, \end{cases}$$

  and is therefore bounded in a neighborhood of $u = 0$. Consequently, it remains to investigate the behavior of the density as $u \to 1^-$ to determine whether it is bounded on $(0,1)$.

  Next, setting $t = -\ln(1-u)$, we can rewrite the density as

$$f_{U_{r,\tilde{\gamma}_r}}(u) = \frac{k^r}{(r-1)!} e^{-(k-1)t} t^{r-1}, \qquad t \to \infty.$$

  If $k > 1$, the exponential decay dominates the polynomial term, and therefore

$$\lim_{u\to 1^-} f_{U_{r,\tilde{\gamma}_r}}(u) = 0.$$

  Since the density is continuous on $(0,1)$ and has finite limits at both endpoints, it is bounded.

  If $k = 1$, the density reduces to

$$f_{U_{r,\tilde{\gamma}_r}}(u) = \frac{[-\ln(1-u)]^{r-1}}{(r-1)!}.$$

Hence,

$$\lim_{u\to 1^-} f_{U_{r,\tilde{\gamma}_r}}(u) = \begin{cases} 1, & r = 1, \\ +\infty, & r > 1. \end{cases}$$

□

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